

Mehil B. Shah

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Research

My research lies at the intersection of Software Engineering (SE) and Artificial Intelligence (AI), with a focus on improving the reliability of AI-powered software systems. My research goals are to: (a) advance the empirical understanding of deep learning bugs and their unique characteristics; (b) develop practical, automated solutions for bug reproduction and fault localization in intelligent systems; (c) bridge traditional software engineering practices and modern AI development workflows through tool building and empirical studies; (d) establish foundations for robust, maintainable agentic AI systems that can be reliably debugged and improved; and (e) train future researchers and practitioners in SE4AI through mentorship, collaboration, and knowledge dissemination.

Education

2023–

Doctor of Philosophy, Computer Science

Dalhousie University, Halifax, NS, Canada

Advisors: Dr. Masud Rahman & Dr. Foutse Khomh

CGPA: 4.25/4.30

Thesis: Automated Reproduction of Deep Learning Bugs Leveraging Agentic AI (tentative)

2016–2020

Bachelor of Technology, Computer Science and Engineering

Manipal University Jaipur, Jaipur, India

CGPA: 9.56/10.00

Thesis: NPQuant: A Robust Quantum-Inspired Computation Algorithm as an Efficient Solution to NP-Complete Problems

Publications (8)

ICSE 26

M. B. Shah, M. Masudur Rahman, and F. Khomh, “Imitation Game: Reproducing Deep Learning Bugs Leveraging an Intelligent Agent”, In Proceedings of *The 48th IEEE/ACM International Conference on Software Engineering (ICSE)*, Rio de Janeiro, Brazil, 2026. (Acceptance Rate: 21.9%)

ACM Artifact Badges: Available · Functional · Reusable

ICSE 26

M. Masudur Rahman and **M. B. Shah**, “Be a Partner, Not a Bystander in Software Engineering Practice: Bridging the Gaps between Academia and Industry”, Future of Software Engineering Track, In Proceedings of *The 48th IEEE/ACM International Conference on Software Engineering (ICSE)*, Rio de Janeiro, Brazil, 2026.

EMSE 25
FSE 25

M. B. Shah, M. Masudur Rahman, and F. Khomh, “Towards Enhancing the Reproducibility of Deep Learning Bugs: An Empirical Study”, *Empirical Software Engineering Journal (EMSE)*, vol. 30, no. 1, article 23, 2025. (Impact Factor: 4.5) Journal-First presentation, Trondheim, Norway.

EMSE 25
ICSE 26

M. B. Shah, M. Masudur Rahman, and F. Khomh, “Towards Understanding the Impact of Data Bugs on Deep Learning Models in Software Engineering”, *Empirical Software Engineering Journal (EMSE)*, vol. 30, article 168, 2025. (Impact Factor: 4.5) Journal-First presentation, Rio de Janeiro, Brazil.

EMSE 25
ICSE 26

S. Jahan, **M. B. Shah**, and M. Masudur Rahman, “Towards Understanding the Challenges of Bug

Localization in Deep Learning Systems”, *Empirical Software Engineering Journal (EMSE)*, 2025. (Impact Factor: 4.5) Journal-First presentation, Rio de Janeiro, Brazil.

- ICSE 25 S. Jahan, **M. B. Shah**, P. Mahbub, and M. Masudur Rahman, “Improved Detection and Diagnosis of Faults in Deep Neural Networks Using Hierarchical and Explainable Classification”, In Proceedings of *The 47th IEEE/ACM International Conference on Software Engineering (ICSE)*, Ottawa, Canada, 2025. (Acceptance Rate: 21.3%)
- ICCWS 24 B. Devi, **M. B. Shah**, V. G. Shankar, and G. Sharma, “NPQuant: A Robust Quantum-Inspired Computation Algorithm as an Efficient Solution to NP-Complete Problems”, In Proceedings of *The International Conference on Cyber Warfare, Security and Space Computing*, pp. 302–313, 2024.
- ISCON 19 **M. B. Shah**, M. Kaistha, and Y. Gupta, “Student Performance Assessment and Prediction System Using Machine Learning”, In Proceedings of *The 4th International Conference on Information Systems and Computer Networks (ISCON)*, pp. 274–279, 2019.
- UNDER REVIEW & PRE-SUBMISSION
- TOSEM **M. B. Shah**, M. M. Morovati, M. Masudur Rahman, and F. Khomh, “Characterizing Faults in Agentic AI: A Taxonomy of Types, Symptoms, and Root Causes”, Under Review in *ACM Transactions on Software Engineering and Methodology (TOSEM)*. (Impact Factor: 6.6)
- ICSE **M. B. Shah**, M. M. Morovati, M. Masudur Rahman, and F. Khomh, “Beyond Scripts: Reproducing Deep Learning Bugs by Reconstructing their Execution Contexts”, Under Review in *The 49th IEEE/ACM International Conference on Software Engineering (ICSE)*.
- ASE **M. B. Shah**, M. Saad, M. Masudur Rahman, and F. Khomh, “Grounded or Guessing? Probing LLM World Knowledge Through Code Execution”, Pre-submission to the *Journal-Ahead Workshop at the ACM/IEEE International Conference on Automated Software Engineering (ASE)*.

Research Grants (2)

- 2026 **Mitacs Lab2Market Validate** (Awarded)
Title: RepGen: Automated Deep Learning Bug Reproduction (in partnership with Springboard Atlantic Inc.)
Role: Collaborator (PI: Dr. Masud Rahman)
Amount: \$10,000 CAD
- 2025 **Dalhousie Research Equipment Fund** (Awarded)
Title: Automated Debugging for AI-Powered Software
Role: Collaborator (PI: Dr. Masud Rahman)
Amount: \$15,000 CAD

Scholarships & Fellowships (5)

- 2026 **NSERC Canada Graduate Research Scholarship – Doctoral (CGRS-D):** Awarded by the Natural Sciences and Engineering Research Council of Canada for the proposal “Reproducibility of Bugs in Agentic AI Systems”; valued at **\$120,000 CAD** over three years.
- 2025 **Performance Top-up:** Awarded \$1,000 CAD in Winter 2025 for strong academic performance.
- 2023 **Dalhousie PhD Scholarship:** Fully-funded admission (fees and living expenses) to the Faculty of Computer Science. Total value: \$108,000 CAD (4 years).

- 2016–2020 **TMA Pai Merit Scholarship:** Awarded a 50% fee waiver ($\approx$ \$10,000 CAD) for the entire tenure of undergraduate studies (Rank 4868 in qualifying exams).
- 2016 **Bhukhanwala Scholarship:** Awarded INR 52,000 ($\approx$ \$800 CAD) for securing the highest grades in Class 12 examinations.

Awards & Honours (8)

- 2025 **Distinguished Reviewer Award (Shadow PC, ICSE 2026):** Recognized for delivering high-quality, constructive, and timely reviews for the Shadow Track of the 48th International Conference on Software Engineering.
- 2025 **Student Volunteer Award (ICSE 2025):** Awarded complimentary registration valued at \$2,000 CAD to the 47th International Conference on Software Engineering, including access to co-located conferences and workshops.
- 2025 **ACM SIGSOFT CAPS Award:** Awarded \$800 USD to support travel and presentation of accepted research at the ACM International Conference on the Foundations of Software Engineering (FSE 2025) in Trondheim, Norway.
- 2025 **Best Paper Presentation Award:** Awarded at CSER 2025, Halifax, Canada for the poster “Towards Enhancing the Reproducibility of Deep Learning Bugs”.
- 2022 **Client Appreciation Award:** Awarded by Accenture for Client Value Creation and Outstanding Dependability.
- 2020 **Fast-track Promotion:** Promoted to Analyst at Accenture for exemplary work and outstanding deliverables.
- 2018 **First Place (Codeathon):** Secured first place at the competitive programming competition organized by IEEE MUJ.
- 2016 **Second Place (TopCoder Hackathon):** Awarded for developing an innovative smart garbage collection system.

Industrial Experience

- 2026 **Mitacs - Springboard Atlantic Inc.,** Halifax, NS
Business Strategy Intern (Lab2Market Validate)
 - Led commercial validation of RepGen through customer-discovery interviews with ML and AI practitioners.
 - Assessed market segments, value proposition, and adoption constraints for translating the prototype into a product.
- 2020–2022 **Accenture,** Pune, India
Senior Software Engineer - Backend & DevOps
 - Built and maintained backend microservices and REST APIs for banking products using Java, Spring Boot, Node.js, and NestJS.
 - Ran monthly release cycles and deployments via Jenkins and OpenShift; co-developed an OTP system with Node.js and Hazelcast.
 - Contributed to proprietary low-code platforms and modernized legacy Angular applications.
- 2020 **Indian Space Research Organisation (ISRO),** India
Research Intern

- Developed a Quantum Approximate Optimization Algorithm (QAOA) to predict MaxCut values from graph structural features, achieving 91.7% prediction accuracy.
- Published the work in the *International Conference on Cyber Warfare and Security (ICCWS 2024)*.

2019

School of Information Sciences, Manipal University, Jaipur, India

Full-Stack Development Intern

- Completed full-stack development training and built a capstone project using Angular and Spring Boot.

2018

Motilal Oswal Financial Services, India

Financial Analytics Intern

- Implemented real-time technical analysis of stocks in Python, applying core technical-indicator concepts from financial markets.

Talks, Posters, & Demos (8)

2026

“Imitation Game: Reproducing Deep Learning Bugs Leveraging an Intelligent Agent”, ICSE, Rio de Janeiro, Brazil.

2026

“Towards Understanding the Impact of Data Bugs on Deep Learning Models in Software Engineering”, ICSE (Journal-First), Rio de Janeiro, Brazil.

2025

“Towards Enhancing the Reproducibility of Deep Learning Bugs: An Empirical Study”, **Poster Presentation**, CSER, Halifax, Canada. (**Best Paper Presentation Award**)

2025

“Towards Enhancing the Reproducibility of Deep Learning Bugs: An Empirical Study”, FSE (Journal-First), Trondheim, Norway.

2024

“Towards Enhancing the Reproducibility of Deep Learning Bugs: An Empirical Study”, **Poster Presentation**, SEMLA, Montreal, Canada.

2024

“NPQuant: A Robust Quantum-Inspired Computation Algorithm as an Efficient Solution to NP-Complete Problems”, International Conference on Cyber Warfare, Security and Space Computing.

2023

“How do Software Bugs Affect the Siamese Neural Network? An Empirical Study”, Research Day Presentation – Systems Cluster (Dalhousie University), Halifax, Canada.

2019

“Student Performance Assessment and Prediction System Using Machine Learning”, ISCON, Mathura, India.

Developed Software Tools & Prototypes (2)

2026

RepGen (*deep-learning bug reproduction*)

An LLM-based agentic system that reproduces deep learning bugs through an iterative generate–validate–refine pipeline over bug reports, repository context, and execution feedback. Reproduced 80% of 106 real-world bugs, validated by 27 practitioners.

ICSE 2026 · [Replication package](#) · [Artifacts: Available](#) · [Functional](#) · [Reusable](#)

2025

DEFault (*fault localization for DNNs*)

A hierarchical fault detection and diagnosis framework for deep neural networks, combining two-stage classification with explainable AI to localize faults at component level with 94% recall — up to 11.54% above prior techniques.

ICSE 2025 · [Replication package](#) · [Artifacts: Available](#)

Teaching

My teaching centers on creating an inclusive, engaging learning environment where students develop both technical competence and critical thinking skills, guided by three core principles: **accessibility**, **active engagement**, and **practical application**. Through weekly tutorials, office hours, and hands-on lab exercises that mirror industry challenges, I help students learn not just how to write code, but how to design, test, and maintain robust software systems. A full teaching statement is available upon request.

Summary: Served as Head TA, TA, and Marker across 16 course offerings at Dalhousie University, supporting over 1,500 students in software engineering, advanced software development, data science, and software project courses.

Responsibilities: Led teams of 3–5 TAs; ran weekly tutorials and labs for 100+ students per term; designed assignments and lab exercises; graded with detailed feedback; held office hours; and mentored junior TAs on pedagogical best practices.

TERM	COURSE	ROLE	ENROL.
Su 2026	CSCI 3130 – Software Eng.	Head TA	76
W 2026	CSCI 3130 – Software Eng.	Head TA	141
F 2025	CSCI 3130 – Software Eng.	Head TA	118
Su 2025	CSCI 3130 – Software Eng.	Head TA	42
Su 2024	CSCI x691 – Project Courses	Head TA	97
W 2025	CSCI 3130 – Software Eng.	TA	134
W 2025	CSCI 5308 – Adv. Software Dev.	TA	46
F 2024	CSCI 3130 – Software Eng.	TA	121
F 2024	CSCI 5308 – Adv. Software Dev.	TA	82
F 2024	CSCI x691 – Project Courses	TA	104
W 2024	CSCI 5308 – Adv. Software Dev.	TA	79
F 2023	CSCI x691 – Project Courses	TA	98
Su 2023	CSCI 3130 – Software Eng.	TA	37
W 2023	CSCI 3130 – Software Eng.	TA	143
W 2023	CSCI x691 – Project Courses	TA	99
Su 2024	CSCI 6409 – Process of Data Sci.	Marker	83

Terms: Su = Summer, F = Fall, W = Winter.

Journal Reviewer

2026	IEEE Transactions on Software Engineering (TSE)
2026	IEEE Transactions on Reliability
2026	Journal of Systems and Software (JSS)
2026	Empirical Software Engineering (EMSE)
2025	ACM Transactions on Software Engineering and Methodology (TOSEM)
2025	IEEE Transactions on Computers
2025	Information and Software Technology (IST)

Organizing Committee (3)

- 2026 **Web Co-Chair:** The ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (**FSE 2026**).
- 2025 **Web Co-Chair:** The 34th International Joint Conference on Artificial Intelligence (**IJCAI 2025**).
- 2025 **Web Co-Chair:** The 32nd IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER 2025).

Program Committee (12)

- 2027 **PC Member:** The 49th IEEE/ACM International Conference on Software Engineering, Research Track (**ICSE 2027**).
- 2026 **Shadow PC Member (Research Track):** The 48th IEEE/ACM International Conference on Software Engineering (**ICSE 2026**) (**Distinguished Reviewer Award**).
- 2026 **PC Member:** The ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (**FSE 2026**), Ideas, Visions and Reflections Track.
- 2026 **PC Member:** The ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (**FSE 2026**), Artifact Evaluation Track.
- 2026 **PC Member:** The 3rd ACM International Conference on AI-powered Software (AIware 2026), Research Track.
- 2026 **PC Member:** The 5th Deep Learning for Code Workshop at ICML 2026, Main Track.
- 2026 **PC Member:** The 33rd IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER 2026), Short Papers & Posters Track.
- 2026 **PC Member:** The 33rd IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER 2026), Industry Track.
- 2026 **PC Member:** The 42nd IEEE International Conference on Software Maintenance and Evolution (ICSME 2026), Tool Demonstration & Showcase Track.
- 2025 **PC Member:** The 4th Deep Learning for Code Workshop at NeurIPS 2025 (DL4C 2025).
- 2025 **PC Member:** The 35th IEEE International Conference on Collaborative Advances in Software and Computing (CASCON 2025), Artifact Evaluation Track.
- 2024 **Junior PC Member:** The 21st International Conference on Mining Software Repositories (MSR 2024).
- 2023–26 **Sub-reviewer:** ICSE, ASE, ESEC/FSE, SANER, ICSME, MSR.
- 2023–26 **Sub-reviewer:** IEEE Transactions on Software Engineering (TSE), ACM Transactions on Software Engineering and Methodology (TOSEM), Journal of Systems and Software (JSS).

Student Volunteering

- 2026 **ICSE 2026** — 48th IEEE/ACM International Conference on Software Engineering, Rio de Janeiro, Brazil.
- 2025 **ICSE 2025** — 47th IEEE/ACM International Conference on Software Engineering, Ottawa, Canada.
- 2025 **CSER 2025** — Consortium for Software Engineering Research, Halifax, Canada.
- Assisted with session management, registration, and logistics to support technical sessions and workshops.*

Leadership & Service

VP Finance and Operations, DOGS — budget planning, operational logistics, and event coordination.

Director of Science and Computer Science, DAGS — organized seminars and competitions and mentored student projects.

Campaigns & Outreach Coordinator, Gramiksha — led volunteer recruitment and digital outreach (youth-focused NGO).

Professional Memberships: ACM SIGSOFT Membership (2025 – Present); IEEE Membership (2025 – Present).

Media Mentions & Publicity

Sep 2025

Bugs and Artificial Intelligence: What is Happening in the Field of Software Engineering?
Futurum Careers

Featured article highlighting the RAISE Lab's research on the intersection of AI and software reliability. Discusses the critical challenges of deep learning bugs and the development of automated debugging agents like *RepGen* and *DEFault* to address them.

Tool & Technology Experience

Programming Languages: C, C++, Java, Python, JavaScript, HTML, CSS.

Frontend Technologies: React, Next.js, Redux, Angular, Bootstrap, Tailwind CSS.

Backend Frameworks & Libraries: Node.js, NestJS, Express, Spring Boot, JUnit, Mockito.

AI, ML & Deep Learning: PyTorch, TensorFlow, Keras, Hugging Face Transformers, CUDA, Reinforcement Learning.

Agentic AI & LLM Frameworks: LangChain, AutoGen, OpenAI APIs, Tool-Calling Agents, Multi-Agent Systems, Prompt Engineering.

Data & Databases: MySQL, MongoDB, Pandas, NumPy.

MLOps & Experimentation: MLflow, Model Versioning, Reproducible Training Pipelines.

DevOps & Infrastructure: Linux, Git, NPM, Maven, TeamCity, Jenkins, Docker, Kubernetes, CI/CD Pipelines.

Professional References

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